Stand-alone PV Systems

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LECTURE OBJECTIVES

- Sizing approach
- System availability
- Hybrid systems
- System sizing approach
- System sizing example
- Tables for sizing components

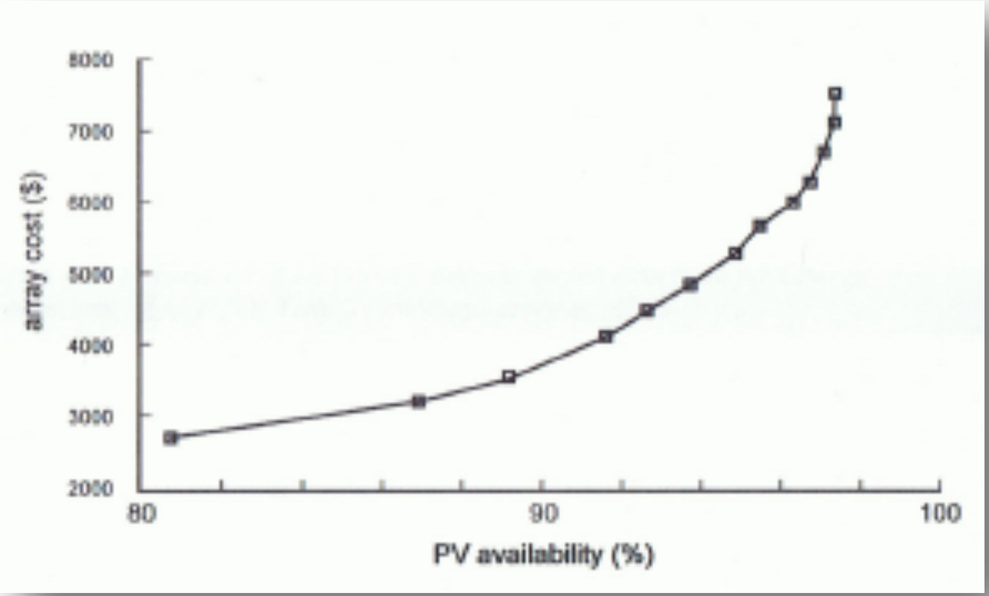
SIZING PROCESS

1. Estimating the electrical load
2. Sizing and specifying batteries
3. Sizing and specifying an array
4. Specifying a controller
5. Sizing and specifying an inverter
6. Sizing system wiring
   - PV Box to Battery
   - PV Box to Controller
   - Controller to Battery
   - Battery to DC Load
   - Battery to Inverter

- Sizing
- Orientation
- Mounting options
- Modules
- Wiring
- Controllers
- Battery storage
- Loads

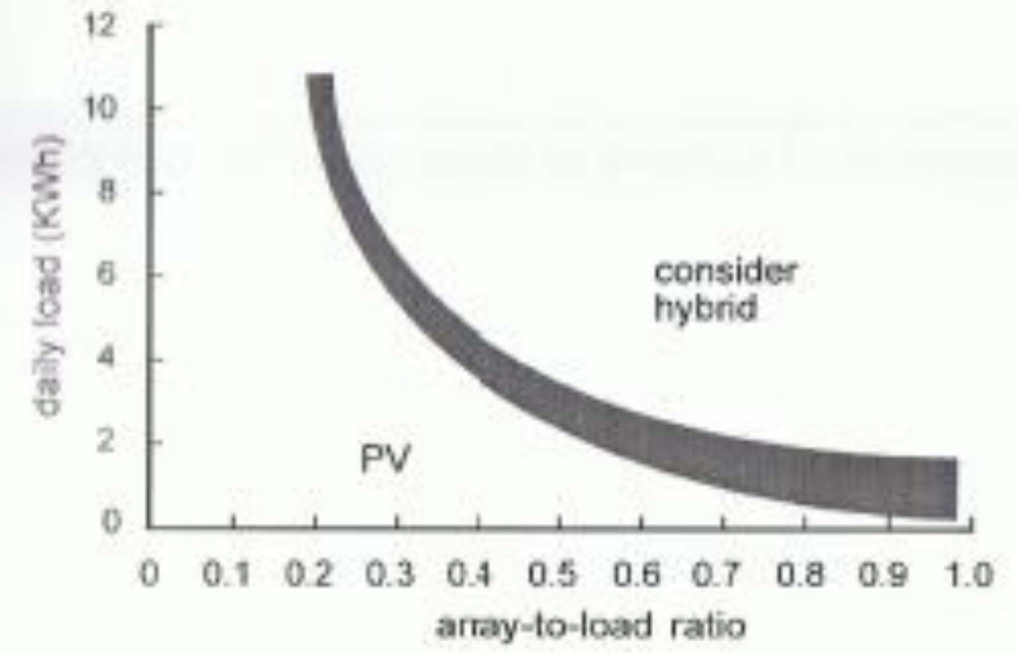
SYSTEM AVAILABILITY

- Percentage of time a system is delivering energy
- Critical applications: e.g., telecommunication
- Non-critical: e.g., cathodic protection
- Low availability: grid-connected system
- Causes for system being not available: weather, failures, maintenance, etc.



HYBRID SYSTEMS

- PV supplies some of the load and diesel or petrol generators are used as backups – allows PV system to be designed with low availability → cost savings
- Hybrid indicator: "array to load ratio" is the array power rating in peak watts divided by the load in watt-hours per day (i.e., W_p/Wh)



SYSTEM SIZING APPROACH

1. Load determination:
    1. Nominal system voltage
    2. Range of voltages tolerated by load
    3. Average load per day
    4. Load profile throughout the year
    5. Example: V = 24 V, average load 100 W, current 4.17 A
2. Select battery capacity:
    1. Conservative approach for certain applications such as telecommunication (e.g., 15 days of battery storage and high availability).
    2. Example: 4.17 A x 24 h x 15 days = 1500 Ah
3. First approximation of tilt angle: typically 20° greater than the latitude
4. Insolation

SYSTEM SIZING APPROACH

5. First approximation of array size: 5 times the average load current
    1. The sun does not shine at night
    2. Reduced light intensity during mornings, afternoons and cloudy weather
    3. The batteries have a limited charging efficiency
    4. Self-discharge of the batteries
    5. Partial blockage of light by dust and debris
6. Optimizing array tilt angle: minimizing depth-of-discharge of the batteries
7. Optimizing array size
8. Summarizing the design

SYSTEM SIZING EXAMPLE

- Stand-alone Photovoltaic System:
  - Lights
    - One 20 W fluorescent light used 4 h per day
    - One 10 W incandescent light used 4 h per day
    - Three 25 W incandescent lights used 4 h per day
  - No inverter
  - Battery
    - Room temperature operation
    - Five days of autonomy
    - Maximum depth of discharge 50%
    - 12 V batteries rated at 100 Ah (over 16 hours) made by Company X

SYSTEM SIZING EXERCISE, cont.

- Array
  - Pole mounted at a 55° tilt angle at 40° N latitude
  - Design for winter conditions
  - 12 V module with a current rating of 2.23 A (at 1000 W/m^2 and 25°C) and I_{sc} = 2.4 A
- Controller
  - Charging capacity rated at 9 A
  - Load switching rated at 10 A
- Wiring
  - 28 feet from the array to controller
  - 2 feet from the control to battery
  - 10 feet from the controller to DC load
  - Wire type: THWN in conduit, 2% voltage drop for all circuits

STAND-ALONE ELECTRICAL LOAD WORKSHEET

INDIVIDUAL LOADS	Qty x	Volts x	Amps =	Watts		x Use	x Use ÷	7 =	Watt Hours	
				AC	DC	hours /day	days/ week		AC	DC
								7		
								7		
								7		
								7		
								7		
AC Total Connected Watts: ______					AC Average Daily Load ______					
DC Total Connected Watts: _____					DC Average Daily Load _____					

INVERTER SIZING WORKSHEET

INVERTER SIZING WORKSHEET						
AC Total Connected W	÷	DC System Voltage	=	Maximum DC Amps cont.	Estimated Surge Watts	Listed features
	÷		=			
Inverter specification		Make		Model:		

BATTERY SIZING WORKSHEET

BATTERY SIZING WORKSHEET										
AC Average Daily Load	÷	Inverter Efficiency	+	DC Average Daily Load	÷	DC System Voltage	=	Average Amp-hour/day		
[(	÷	)	+	]	÷		=			
Average Amp-hour/day	X	Days of Autonomy	÷	Discharge Limit	÷	Battery AH Capacity	=	Batteries in parallel		
	X		÷		÷		=			
DC System Voltage	÷	Battery Voltage	=	Batteries in Series	X	Batteries in parallel	=	Total Batteries		
	÷		=		X		=			
Battery specification		Make				Model:				

ARRAY SIZING WORKSHEET								
Average Amp-h/day	÷	Battery Efficiency	÷	Peak Sun Hrs/day	=	Average hours/day		
	÷		÷		=			
Average Peak Amps	÷	Peak Amps/module	=	Modules in Parallel		Module Short Circuit current		
	÷		=					
DC System Voltage	÷	Nominal Module Voltage	=	Modules in Series	X	Modules in parallel	=	Total Modules
	÷		=		X		=	
Battery specification		Make		Model:				

CONTROLLER SIZING WORKSHEET

CONTROLLER SIZING WORKSHEET								
Module Short Circuit current	X	Modules in parallel	X	1.25	=	Array Short Circuit Amps	Controller Array Amps	Desired Features
	X		X	1.25	=			
DC Total Connected W	÷	DC System Voltage	=	Maximum DC Load Amps			Controller Load Amps	
	÷		=					
Controller specification			Make		Model:			

WIRE SIZING WORKSHEET

WIRE SIZING WORKSHEET										
A. NEC REQUIREMENT										
I_{SC} of modules	X	# Modules in parallel	=	Total Amps	X	1.25	X	1.25	=	NEC required amps
	X		=		X	1.25	X	1.25	=	
Amperage satisfying NEC =				Wire size (from table) =						

B. VOLTAGE DROP REQUIREMENTS

System Voltage ___________ Total Amps _______________

One Way Distance ____________ Voltage Drop (%) _____________

Wire size (from voltage drop tables) __________________

Is this equal to or greater than the size wired needed for safety? ________________

If yes, this is your answer If no, use the wire size from A

REFLECTIONS AND OBSERVATIONS

Pause:
What did you take from this section ?
What's missing ?
What needs additional class explanation ?

Activity:

- Do you feel you can calculate the system sizing?
- Share with the group
- Report to rest of the class

ESSENTIAL KNOWLEDGE QUESTIONS

- What is a “hybrid” system when considering photovoltaics? What are the applications best suited for a hybrid system?

READING MATERIAL/PREPARATION

- "Energy analysis – solar technology analysis models and tools", National Renewable Energy Laboratory (2004). (www.nrel.gov/analysis/analysis_tools_tech_sol.html)
- "Stand alone photovoltaic systems - a handbook of recommended design practices", Sandia National Laboratory (1991). (www.sandia.gov/pv/docs/programmatic.html)
- http://www.advancepower.net/advcalc.htm